Topic 3: Solved Problems

Full step-by-step analytical resolutions for the 6 official problem set exercises on Diffusion in Solids and Mass Transport, formatted with rigorous 4-phase pedagogical derivations.

OFFICIAL UC3M ERROR FUNCTION REFERENCE TABLE [SESSION 5 SLIDE 21]

z	$\mathrm{erf}(z)$	z	$\mathrm{erf}(z)$	z	$\mathrm{erf}(z)$	z	$\mathrm{erf}(z)$
0.00	0.0000	0.25	0.2763	0.70	0.6778	1.30	0.9340
0.01	0.0113	0.30	0.3286	0.75	0.7112	1.40	0.9523
0.02	0.0226	0.35	0.3794	0.80	0.7421	1.50	0.9661
0.03	0.0338	0.40	0.4284	0.85	0.7707	1.60	0.9763
0.04	0.0451	0.45	0.4755	0.90	0.7969	1.70	0.9838
0.05	0.0564	0.50	0.5205	0.95	0.8209	1.80	0.9891
0.10	0.1125	0.55	0.5633	1.00	0.8427	1.90	0.9928
0.15	0.1680	0.60	0.6039	1.10	0.8802	2.00	0.9953
0.20	0.2227	0.65	0.6420	1.20	0.9103		

`Problem 1` `Problems T3_Diffusion.pdf #1` `Fick's 2nd Law · erf(z)`

Cementation of Grade 1018 Steel Gear

`Original statement · Problems T3_Diffusion.pdf`

1. In a steel gear of grade 1018 (0.18 wt% C*) a cementation process is carried out at* $927^{\circ}\mathrm{C}$*. Calculate the time required in order to increase the* C *content up to* 0.3 wt% *at* 0.6 mm *below the surface of the gear. Assume that the* C *content in the surface is* 1 wt%*. Data:* $D_{\mathrm{C\ in\ }\gamma\text{-Fe}} = 1.28 \times 10^{-11}\ \mathrm{m}^2/\mathrm{s}$.

`Official Solution: t = 6636 s (1.84 h)`

PHASE 1 · HYPOTHESES & PARAMETERS

- **Semi-Infinite Medium:** The penetration depth $x = 0.6$ mm is small compared with the size of the part, allowing one-dimensional semi-infinite modeling [Session 5 Slide 20].
- **Constant Surface Concentration:** The carburizing atmosphere maintains $C(0,t) = C_s = 1.0$ wt% for $t > 0$.
- **Homogeneous Initial State:** The AISI 1018 steel core has initial carbon concentration $C(x,0) = C_0 = 0.18$ wt%.
- **Parameters:** $C_0 = 0.18$ wt%, $C_s = 1.00$ wt%, $C_x = 0.30$ wt%, $x = 0.6$ mm $= 6.0 \times 10^{-4}$ m, $D = 1.28 \times 10^{-11}$ m^2/s at $T = 927^\circ$C.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

In non-steady-state diffusion into a semi-infinite solid with constant surface composition, Fick's Second Law yields [Session 5 Slide 20]:

$$\frac{C_s - C_x}{C_s - C_0} = \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right) = \operatorname{erf}(z)$$

We evaluate the concentration ratio, locate the corresponding argument z in the official error function table via exact linear interpolation, and solve for time t:

$$z = \frac{x}{2\sqrt{Dt}} \implies t = \frac{x^2}{4 \cdot z^2 \cdot D}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Concentration Ratio:

$$\frac{C_s - C_x}{C_s - C_0} = \frac{1.00 - 0.30}{1.00 - 0.18} = \frac{0.70}{0.82} \approx 0.8536585 \implies \operatorname{erf}(z) = 0.8536585$$

2. Linear Interpolation in Error Function Table [Slide 21]:

- For $z_1 = 1.00 \implies \operatorname{erf}(z_1) = 0.8427$
- For $z_2 = 1.10 \implies \operatorname{erf}(z_2) = 0.8802$

$$\frac{z-1.00}{1.10-1.00}=\frac{0.8536585-0.8427}{0.8802-0.8427}\implies\frac{z-1.00}{0.10}=\frac{0.0109585}{0.0375}\approx 0.292227$$

$$z=1.00+0.10\times 0.292227=1.02922\implies z^2=(1.02922)^2\approx 1.05930$$

3. Time of Carburization:

$$x^2=(6.0\times10^{-4}\ \mathrm{m})^2=3.60\times10^{-7}\ \mathrm{m}^2$$

$$4\cdot z^2\cdot D=4\times1.05930\times(1.28\times10^{-11}\ \mathrm{m}^2/\mathrm{s})=5.4236\times10^{-11}\ \mathrm{m}^2/\mathrm{s}$$

$$t=\frac{3.60\times10^{-7}\ \mathrm{m}^2}{5.4236\times10^{-11}\ \mathrm{m}^2/\mathrm{s}}\approx\mathbf{6638}\ \mathrm{s}=\mathbf{1.844}\ \mathrm{h}$$

Discrepancy with the official solution. The official key gives $t=6636$ s. Linear interpolation in the table gives $z=1.0292$ and $t=6637.6$ s ≈ 6638 s (the exact inverse error function, $z=1.0271$, would give 6665 s because the table is linear between 1.00 and 1.10). The key is 0.02% below the interpolated value; this is a rounding difference, and the interpolated value is reported.

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[t]=\frac{\mathrm{m}^2}{\mathrm{m}^2/\mathrm{s}}=\mathrm{s}$. **Physical Insight:** About 1.84 hours at 927°C bring the carbon content at 0.6 mm to 0.30 wt%. Carbon is small, so it diffuses interstitially [Slide 13]; in the surface layer it blocks the movement of dislocations and hardens the steel for wear resistance, while the core stays tough, and the excess carbon leaves residual compressive stresses [Slide 30].

Problem 2 | Problems T3_Diffusion.pdf #2 | Arrhenius · Semiconductor Doping

Diffusion of Aluminum in Monocrystalline Silicon

Original statement · Problems T3_Diffusion.pdf

2.- Al *can diffuse in a* Si *monocrystal. Calculate the temperature at which the diffusion coefficient will have a value of* $10^{-14}\ \mathrm{m^2/s}$*? Data:* $E_D = 73\ \mathrm{kcal/mol}$*;* $D_0 = 1.55 \times 10^{-4}\ \mathrm{m^2/s}$ *and* $R = 1.987\ \mathrm{cal/mol \cdot K}$. Official Solution: T = 1566 K (1293 °C)

PHASE 1 · HYPOTHESES & PARAMETERS

- **Volume Diffusion in a Single Crystal:** A single crystal has no grain boundaries, so diffusion takes place through the volume, the slowest path ($D_{surface} > D_{grain\ boundary} > D_{volume}$) [Slide 25].
- **Arrhenius Behavior:** The temperature dependence of D obeys Arrhenius kinetics with constant D_0 and E_D [Slide 26].
- **Parameters:** $D = 10^{-14}\ \mathrm{m^2/s}$, $D_0 = 1.55 \times 10^{-4}\ \mathrm{m^2/s}$, $E_D = 73\,000\ \mathrm{cal/mol}$, $R = 1.987\ \mathrm{cal/mol \cdot K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

Starting from the Arrhenius equation [Session 5 Slide 26]:

$$D = D_0 \exp\left(-\frac{E_D}{RT}\right) \implies \ln\left(\frac{D_0}{D}\right) = \frac{E_D}{RT}$$

Isolating absolute temperature T:

$$T = \frac{E_D}{R \cdot \ln(D_0/D)}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Diffusivity Ratio:

$$\frac{D_0}{D} = \frac{1.55 \times 10^{-4}\ \mathrm{m^2/s}}{1.0 \times 10^{-14}\ \mathrm{m^2/s}} = 1.55 \times 10^{10}$$

2. Natural Logarithm:

$$\ln(1.55 \times 10^{10}) = \ln(1.55) + 10 \cdot \ln(10) = 0.438255 + 23.025851 = 23.464106$$

3. Absolute Temperature Calculation:

$$T = \frac{73\,000\ \mathrm{cal/mol}}{(1.987\ \mathrm{cal/mol \cdot K}) \times (23.464106)} = \frac{73\,000}{46.62318} \approx \mathbf{1565.74}\ \mathrm{K} \approx \mathbf{1566}\ \mathrm{K}$$

Conversion to Celsius:

$$T = 1565.74 - 273.15 = \mathbf{1292.59}^\circ\mathrm{C} \approx \mathbf{1293}^\circ\mathrm{C}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[T] = \frac{\mathrm{cal/mol}}{(\mathrm{cal/mol \cdot K})} = \mathrm{K}$. **Physical Insight:** A diffusivity of $10^{-14}\ \mathrm{m^2/s}$ needs $1293^\circ\mathrm{C}$: D depends exponentially on T, and the high activation energy ($73\ \mathrm{kcal/mol} \approx 305\ \mathrm{kJ/mol}$) means the atoms need a lot of thermal energy to jump [Slides 4, 26]. Diffusion of impurities in Si wafers is the process used to fabricate integrated circuits [Slide 29].

Problem 3 | Problems T3_Diffusion.pdf #3 | Two-Point Arrhenius · Activation Energy

Activation Energy and Diffusivity of Carbon in Mild Steel

Original statement · Problems T3_Diffusion.pdf

3.- The diffusivity of carbon in mild steel was measured at two different temperatures:
Data: $R = 8.314\ \text{J/mol} \cdot \text{K}$
• *At* $T_1 = 850^\circ\text{C}$*:* $D_1 = 4.826 \times 10^{-12}\ \text{m}^2/\text{s}$
• *At* $T_2 = 950^\circ\text{C}$*:* $D_2 = 1.805 \times 10^{-11}\ \text{m}^2/\text{s}$
Using these data, calculate:
a) The activation energy for the C *diffusion in mild steel, for the temperature interval of* 850°C *to* 950°C *(Solution:* $E_D = 36\ \text{kcal/mol}$*)*
b) The diffusivity at 1000°C*, assuming that the activation energy does not change at this temperature range. (Solution:* $D = 3.23 \times 10^{-11}\ \text{m}^2/\text{s}$*)*

Official Solution: ED = 36 kcal/mol; D(1000 °C) = 3.23 x 10^-11 m^2/s

PHASE 1 · HYPOTHESES & PARAMETERS

- **Single Arrhenius Law (assumption):** The problem uses one Arrhenius law with constant D_0 and E_D for C in mild steel from 850°C to 1000°C, as part b) states [Session 5 Slide 26].
- **Parameters:** $T_1 = 850^\circ\text{C} = 1123.15\ \text{K}$, $D_1 = 4.826 \times 10^{-12}\ \text{m}^2/\text{s}$ $T_2 = 950^\circ\text{C} = 1223.15\ \text{K}$, $D_2 = 1.805 \times 10^{-11}\ \text{m}^2/\text{s}$ $T_3 = 1000^\circ\text{C} = 1273.15\ \text{K}$, $R = 8.314\ \text{J/mol} \cdot \text{K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

Dividing Arrhenius equations at two temperatures eliminates D_0 [Session 5 Slide 26]:

$$\ln\left(\frac{D_2}{D_1}\right) = \frac{E_D}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \implies E_D = \frac{R \cdot \ln(D_2/D_1)}{\frac{1}{T_1} - \frac{1}{T_2}}$$

For part (b), extrapolating to $T_3 = 1000^\circ\text{C}$:

$$D_3 = D_2 \cdot \exp\left[\frac{E_D}{R}\left(\frac{1}{T_2} - \frac{1}{T_3}\right)\right]$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

Part (a): Activation Energy E_D

$$\frac{D_2}{D_1} = \frac{1.805 \times 10^{-11}}{4.826 \times 10^{-12}} = 3.7401575 \implies \ln(3.7401575) \approx 1.319128$$

$$\frac{1}{T_1} - \frac{1}{T_2} = \frac{1}{1123.15} - \frac{1}{1223.15} = 8.903530 \times 10^{-4} - 8.175612 \times 10^{-4} = 7.27918 \times 10^{-5}\ \mathrm{K}^{-1}$$

$$E_D = \frac{8.314 \times 1.319128}{7.27918 \times 10^{-5}} = 150\,666\ \mathrm{J/mol} = 150.67\ \mathrm{kJ/mol}$$

$$E_D = \frac{150\,666\ \mathrm{J/mol}}{4184\ \mathrm{J/kcal}} \approx \mathbf{36.01}\ \mathrm{kcal/mol} \approx \mathbf{36}\ \mathrm{kcal/mol}$$

Part (b): Diffusivity at $1000^\circ\mathrm{C}$ ($T_3 = 1273.15$ K)

$$\frac{1}{T_2} - \frac{1}{T_3} = \frac{1}{1223.15} - \frac{1}{1273.15} = 8.175612 \times 10^{-4} - 7.854534 \times 10^{-4} = 3.21078 \times 10^{-5}\ \mathrm{K}^{-1}$$

$$\text{Exponent} = \frac{150\,666}{8.314} \times (3.21078 \times 10^{-5}) = 18121.96 \times 3.21078 \times 10^{-5} \approx 0.581857$$

$$D_3 = (1.805 \times 10^{-11}\ \mathrm{m^2/s}) \times e^{0.581857} = (1.805 \times 10^{-11}) \times 1.78936 \approx \mathbf{3.23 \times 10^{-11}}\ \mathbf{m^2/s}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Consistency with Official Data: The official reference table [Session 5 Slide 23] lists $D(\mathrm{C\ in\ Fe\text{-}FCC}) \approx 3 \times 10^{-11}\ \mathrm{m^2/s}$ at $1000^\circ\mathrm{C}$, matching our derived $3.23 \times 10^{-11}\ \mathrm{m^2/s}$; both are of the same order, the slide value being a one-significant-figure figure ($\approx 8\%$ lower). **Mechanism Check:** $E_D \approx 36$ kcal/mol is lower than the activation energy for self-diffusion of iron (57.5 kcal/mol $= 240$ kJ/mol for α-Fe) [Slide 9]. This agrees with carbon diffusing interstitially, which needs less energy than vacancy diffusion [Slide 13].

Problem 4 | Problems T3_Diffusion.pdf #4 | Fick's 1st Law · Ceramic Barrier Transport

Ionic Transport of Nickel Through an MgO Ceramic Plate

Original statement · Problems T3_Diffusion.pdf

4.- A plate of magnesium oxide of $0.1\ \text{cm}$ *thickness separates two metallic blocks, one of* Ni *and one of* Ta*. At* 1400°C Ni *ions diffuse through the* MgO *plate. The diffusion coefficient of* Ni *in* MgO *is* $9 \times 10^{-12}\ \text{cm}^2/\text{s}$ *and the lattice parameter of (fcc)* Ni *at* 1400°C *is* $3.6 \times 10^{-8}\ \text{cm}$*. Determine the time necessary for sufficient ions* Ni^{2+} *ions to pass through the ceramic material so that the thickness of the block of* Ni *is reduced by one micron.*

Official Solution: t = 309 h

PHASE 1 · HYPOTHESES & PARAMETERS

- **Quasi-Steady State:** Because the plate thickness ($\Delta x = 0.1\ \text{cm} = 1000\ \mu\text{m}$) is much larger than the layer consumed ($\Delta h = 1\ \mu\text{m}$), flux through MgO is in steady state [Slide 16].
- **Interfacial Boundary Conditions:** At the Ni/MgO interface, the concentration equals the atomic density of nickel metal: $C_1 = \rho_{\text{at, Ni}} = 4/a^3$ (FCC, as the statement gives). At the MgO/Ta interface, the nickel that arrives is taken as removed at once: $C_2 \approx 0$ (assumption of the model).
- **Parameters:** $\Delta x = 0.1\ \text{cm}$, $D = 9 \times 10^{-12}\ \text{cm}^2/\text{s}$, $a = 3.6 \times 10^{-8}\ \text{cm}$, $\Delta h = 1\ \mu\text{m} = 10^{-4}\ \text{cm}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Atoms of Ni removed per unit area: $N/A = \rho_{\text{at, Ni}} \cdot \Delta h = \frac{4 \cdot \Delta h}{a^3}$.
2. Steady-state flux via Fick's First Law [Session 5 Slide 16]:

$$J = D\frac{\Delta C}{\Delta x} = D\frac{\rho_{\text{at, Ni}}}{\Delta x}$$

3. Conservation of mass equates flux to removal rate:

$$J = \frac{N/A}{t} \implies \frac{\rho_{\text{at, Ni}} \cdot \Delta h}{t} = D\frac{\rho_{\text{at, Ni}}}{\Delta x} \implies t = \frac{\Delta x \cdot \Delta h}{D}$$

The atomic density $\rho_{\text{at, Ni}}$ elegantly cancels out.

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Direct Evaluation with General Formula:

$$\Delta x = 0.1\ \text{cm}, \quad \Delta h = 10^{-4}\ \text{cm}, \quad D = 9 \times 10^{-12}\ \text{cm}^2/\text{s}$$

$$t = \frac{(0.1\ \text{cm}) \times (10^{-4}\ \text{cm})}{9 \times 10^{-12}\ \text{cm}^2/\text{s}} = \frac{10^{-5}\ \text{cm}^2}{9 \times 10^{-12}\ \text{cm}^2/\text{s}} = \frac{1}{9} \times 10^{7}\ \text{s} \approx 1.1111 \times 10^{6}\ \text{s}$$

2. Conversion to Hours:

$$t = \frac{1.1111 \times 10^{6}\ \text{s}}{3600\ \text{s/h}} \approx \mathbf{308.64}\ \text{h} \approx \mathbf{309}\ \text{h}$$

3. Step-by-Step Atomic Confirmation:

$$\rho_{\text{at, Ni}} = \frac{4}{(3.6 \times 10^{-8}\ \text{cm})^3} = 8.5734 \times 10^{22}\ \text{atoms/cm}^3$$

$$N/A = (8.5734 \times 10^{22}) \times (10^{-4}) = 8.5734 \times 10^{18}\ \text{atoms/cm}^2$$

$$J = (9 \times 10^{-12}) \times \frac{8.5734 \times 10^{22}}{0.1} = 7.71605 \times 10^{12}\ \text{atoms/cm}^2 \cdot \text{s}$$

$$t = \frac{8.5734 \times 10^{18}}{7.71605 \times 10^{12}} = 1.1111 \times 10^{6}\ \text{s} \approx \mathbf{309}\ \text{h}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[t] = \frac{\text{cm} \cdot \text{cm}}{\text{cm}^2/\text{s}} = \text{s}$. **Physical Insight:** Nearly 13 days (309 hours) at 1400°C are needed to remove 1 μm of nickel: diffusion is slower for charged species, in compact structures and in materials with high T_m [Slide 27]. At fixed concentrations the flux falls as $1/\Delta x$, so a thicker plate would take proportionally longer [Slide 16].

Problem 5 | Problems T3_Diffusion.pdf #5 | Fick's 2nd Law + Arrhenius Coupled

Cementation Temperature for Steel 1010 in 8 Hours

Original statement · Problems T3_Diffusion.pdf

5.- A 1010 steel, that contains $0.10\ \text{wt\%}\ \text{C}$ *is subjected to cementation process, in order to obtain effective* C *composition of* $0.30\ \text{wt\%}$ *at a depth of* $1.16\ \text{mm}$ *below the surface. Assume that the maximum* C *solubility in the steel surface is reached at moment that the carburizing process is initiated, generating a* C *concentration at the surface of* $1.31\ \text{wt\%}$*. Determine the required cementation temperature for the process to be successfully carried out within an 8 hour time span. Data:* $D_0(\text{C in Fe}) = 16.2\ \text{mm}^2/\text{s}$*;* $E_D = 137\,800\ \text{J/mol}$*;* $R = 8.314\ \text{J/mol}\cdot\text{K}$.

Official Solution: T = 1175 K = 902 °C

PHASE 1 · HYPOTHESES & PARAMETERS

- **Coupled Transient Diffusion:** Combines semi-infinite error function diffusion with Arrhenius thermal kinetics [Session 5 Slides 20, 26].
- **Parameters:** $C_0 = 0.10\ \text{wt\%}$, $C_s = 1.31\ \text{wt\%}$, $C_x = 0.30\ \text{wt\%}$ $x = 1.16\ \text{mm} = 1.16 \times 10^{-3}\ \text{m}$ $t = 8\ \text{h} = 8 \times 3600\ \text{s} = 28\,800\ \text{s}$ $D_0 = 16.2\ \text{mm}^2/\text{s} = 16.2 \times 10^{-6}\ \text{m}^2/\text{s}$ $E_D = 137\,800\ \text{J/mol}$, $R = 8.314\ \text{J/mol}\cdot\text{K}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Evaluate the concentration ratio $\text{erf}(z) = \frac{C_s - C_x}{C_s - C_0}$.
2. Determine z via linear interpolation in the official table [Slide 21].
3. Calculate the required diffusivity: $D = \frac{x^2}{4 \cdot z^2 \cdot t}$.
4. Invert the Arrhenius equation to calculate the processing temperature T:

$$T = \frac{-E_D}{R \cdot \ln(D/D_0)}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Error Function Value:

$$\text{erf}(z) = \frac{1.31 - 0.30}{1.31 - 0.10} = \frac{1.01}{1.21} \approx 0.83471074$$

2. Linear Interpolation in Error Function Table [Slide 21]:

- For $z_1 = 0.95 \implies \text{erf}(z_1) = 0.8209$
- For $z_2 = 1.00 \implies \text{erf}(z_2) = 0.8427$

$$\frac{z - 0.95}{1.00 - 0.95} = \frac{0.83471074 - 0.8209}{0.8427 - 0.8209} \implies \frac{z - 0.95}{0.05} = \frac{0.01381074}{0.0218} \approx 0.633520$$

$$z = 0.95 + 0.05 \times 0.633520 = 0.981676 \implies z^2 = (0.981676)^2 \approx 0.963688$$

3. Diffusivity Calculation:

$$x^2 = (1.16 \times 10^{-3}\ \text{m})^2 = 1.3456 \times 10^{-6}\ \text{m}^2$$

$$4 \cdot z^2 \cdot t = 4 \times 0.963688 \times 28\,800\ \text{s} = 111\,016.86\ \text{s}$$

$$D = \frac{1.3456 \times 10^{-6}\ \text{m}^2}{111\,016.86\ \text{s}} \approx \mathbf{1.21207 \times 10^{-11}}\ \text{m}^2/\text{s}$$

4. Inverting Arrhenius for Temperature T:

$$\frac{D}{D_0} = \frac{1.21207 \times 10^{-11}\ \text{m}^2/\text{s}}{16.2 \times 10^{-6}\ \text{m}^2/\text{s}} = 7.48191 \times 10^{-7} \implies \ln\left(\frac{D}{D_0}\right) = \ln(7.48191 \times 10^{-7}) = 2.01249 - 16.11810 = -14.10561$$

$$T = \frac{-137\,800\ \text{J/mol}}{(8.314\ \text{J/mol}\cdot\text{K}) \times (-14.10561)} = \frac{137\,800}{(8.314)(14.10561)} = \frac{137\,800}{117.275} \approx \mathbf{1175.02}\ \text{K} \approx \mathbf{1175}\ \text{K}$$

$$T = 1175.02 - 273.15 = \mathbf{901.87}^\circ\text{C} \approx \mathbf{902}^\circ\text{C}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Sensitivity to Temperature: To reach $0.30\ \mathrm{wt}\%$ at $1.16\ \mathrm{mm}$ in $8\ \mathrm{h}$ the steel must be carburized at $902^{\circ}\mathrm{C}$. Because D grows exponentially with T, a lower temperature would need a much longer time and a higher one a shorter time [Slide 26].

Problem 6 | Problems T3_Diffusion.pdf #6 | Fick's 1st Law · Membrane Purification

Hydrogen Purification Through a Palladium Membrane

Original statement · Problems T3_Diffusion.pdf

6.- Determine the thickness of a plate of Pd *with transversal area of* $0.2\ \mathrm{m}^2$ *so that it can purify* $1.73 \times 10^{-3}\ \mathrm{kg/h}$ *of hydrogen, if the hydrogen concentration at the side of high pressure of the plate is* $1.5\ \mathrm{kg/m}^3$ *and at the side of low pressure is* $0.3\ \mathrm{kg/m}^3$*. The diffusion coefficient of hydrogen in* Pd *is* $1 \times 10^{-8}\ \mathrm{m}^2/\mathrm{s}$. Official Solution: Δx = 5 mm

PHASE 1 · HYPOTHESES & PARAMETERS

- **Pure Steady State:** Constant external pressures create time-invariant boundary concentrations C_{high} and C_{low} [Session 5 Slide 35].
- **Parameters:** $A = 0.2\ \mathrm{m}^2, \dot{m} = 1.73 \times 10^{-3}\ \mathrm{kg/h}\ C_1 = 1.5\ \mathrm{kg/m}^3, C_2 = 0.3\ \mathrm{kg/m}^3\ D = 1.0 \times 10^{-8}\ \mathrm{m}^2/\mathrm{s}$.

PHASE 2 · FORMULATION & PEDAGOGICAL STRATEGY

1. Compute mass flux density [Session 5 Slide 15]:

$$J = \frac{\dot{m}}{A} \quad \left[\frac{\mathrm{kg}}{\mathrm{m}^2 \cdot \mathrm{s}}\right]$$

2. Apply Fick's First Law across the flat membrane [Slide 16, 35]:

$$J = D\frac{\Delta C}{\Delta x} \implies \Delta x = \frac{D \cdot (C_1 - C_2)}{J}$$

PHASE 3 · STEP-BY-STEP ANALYTICAL DERIVATION

1. Flux Density (J):

$$\frac{\dot{m}}{A} = \frac{1.73 \times 10^{-3}\ \mathrm{kg/h}}{0.2\ \mathrm{m^2}} = 8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h}$$

$$J = \frac{8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h}}{3600\ \mathrm{s/h}} \approx \mathbf{2.4028 \times 10^{-6}}\ \mathrm{kg/m^2 \cdot s}$$

The worked solution stored in the text layer of Official Slide 35 gives $J = 8.65 \times 10^{-3}\ \mathrm{kg/m^2 \cdot h} = 2.4 \times 10^{-6}\ \mathrm{kg/m^2 \cdot s}$ *and* $\Delta x = 5\ \mathrm{mm}$ *(the answer box on the rendered slide appears empty).*

2. Concentration Difference:

$$\Delta C = C_1 - C_2 = 1.5\ \mathrm{kg/m^3} - 0.3\ \mathrm{kg/m^3} = 1.2\ \mathrm{kg/m^3}$$

3. Membrane Thickness (Δx):

$$\Delta x = \frac{(1.0 \times 10^{-8}\ \mathrm{m^2/s}) \times (1.2\ \mathrm{kg/m^3})}{2.4028 \times 10^{-6}\ \mathrm{kg/m^2 \cdot s}} \approx 4.994 \times 10^{-3}\ \mathrm{m} \approx \mathbf{5.0 \times 10^{-3}}\ \mathrm{m} = \mathbf{5}\ \mathrm{mm}$$

PHASE 4 · PHYSICAL & ENGINEERING VERIFICATION

Dimensional Check: $[\Delta x] = \frac{(\mathrm{m^2/s}) \cdot (\mathrm{kg/m^3})}{\mathrm{kg/(m^2 \cdot s)}} = \mathrm{m}$. **Physical Insight:** At fixed concentrations the flux scales as $J \propto 1/\Delta x$ [Slide 16], so 5 mm is the thickness at which steady-state permeation equals the required purified flow. Hydrogen is small with respect to the matrix, so it diffuses interstitially, which is faster [Slide 13]; its diffusivity ($10^{-8}\ \mathrm{m^2/s}$) is far above the values of the diffusivity table [Slide 23]. This is the example of $\mathrm{H_2}$ purification through a Pd layer [Slide 35].